The Complete Integer-Kink Dirac Atlas: Zero Mode, Bound Ladder, Thresholds, and Reflectionlessness

HCS-C352 source-local reconstruction

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Abstract

For every positive integer height, we give a single proof of the complete spectral and scattering atlas of the one-dimensional hyperbolic-tangent Dirac kink. Bounded-perturbation theory fixes the self-adjoint domain; supersymmetric factorization and shape invariance exhaust the simple bound ladder; the unpaired kernel gives exactly one chiral zero mode; and Darboux images of the free zero-momentum state and plane waves settle both threshold resonances and reflectionlessness. Exact ledgers independently check 25 heights and 150 rational complex scattering values, but the all-integer conclusion comes from the analytic descent argument. No priority or arithmetic-target claim is made.

Revision certificate. Round-one threshold-resonance and reflectionless-scattering closure.

1 Frozen operator and theorem

On $L^2(\mathbb{R};\mathbb{C}^2)$ define, for $n \in \mathbb{Z}_{\geq 1}$,

$$A_n = \frac{d}{dx} + n\tanh x, \qquad A_n^* = -\frac{d}{dx} + n\tanh x, \qquad H_n = \begin{pmatrix} 0 & A_n^* \\ A_n & 0 \end{pmatrix}, \tag{1}$$

with domain $H^1(\mathbb{R};\mathbb{C}^2)$. "Upper" and "lower" always refer to the first and second spinor components in (1). A threshold resonance means a nonzero bounded distributional solution which is not in L^2.

Theorem 1 (Integer-kink atlas)**.** *For every integer $n \geq 1$ the following statements hold.*

(i) *H_n is self-adjoint on $H^1(\mathbb{R};\mathbb{C}^2)$ and generates the unitary group e^{-itH_n}.*

(ii) *Its essential spectrum is*

$$\sigma_{\mathrm{ess}}(H_n) = (-\infty, -n] \cup [n, \infty),$$

and the continuous spectral subspace is purely absolutely continuous.

(iii) *The complete point spectrum is*

$$\sigma_{\mathrm{p}}(H_n) = \{0\} \cup \left\{ \pm\sqrt{j(2n-j)} : 1 \leq j \leq n-1 \right\}. \tag{2}$$

Every listed eigenvalue is simple. The zero eigenspace is spanned by the column spinor whose upper component is $\operatorname{sech}^n x$ *and whose lower component is zero.*

(iv) *Both $+n$ and $-n$ are resonances and neither is an L^2 eigenvalue.*

(v) *Both scalar channels of H_n^2, and consequently the Dirac scattering problem, are reflectionless.*

For $n = 1$, (2) contains only the zero mode; this nontrivial boundary case is included in the theorem.

2 Self-adjoint square and shape invariance

The free off-diagonal derivative operator on H^1 is self-adjoint after the fixed signs in (1) are used. Multiplication by the bounded Hermitian matrix with entries $n\tanh x$ is a bounded self-adjoint perturbation. The Kato–Rellich bounded-perturbation theorem proves assertion (i), and Stone's theorem supplies the unitary group.

Direct multiplication, using $(\tanh x)' = \operatorname{sech}^2 x$, gives

$$\begin{aligned} B_n &:= A_n^* A_n = -\frac{d^2}{dx^2} + n^2 - n(n+1)\operatorname{sech}^2 x, \\ C_n &:= A_n A_n^* = -\frac{d^2}{dx^2} + n^2 - n(n-1)\operatorname{sech}^2 x, \\ H_n^2 &= \operatorname{diag}(B_n, C_n). \end{aligned} \tag{3}$$

The key identity is the exact shape invariance

$$C_n = B_{n-1} + 2n - 1. \tag{4}$$

Moreover $B_n A_n^* = A_n^* C_n$ and $A_n B_n = C_n A_n$.

Lemma 1 (Scalar hierarchy)**.** *The point spectrum of B_n consists precisely of the simple values*

$$q_{n,j} = j(2n-j), \qquad j = 0, \dots, n-1. \tag{5}$$

Its essential spectrum is $[n^2, \infty)$ and is purely absolutely continuous. At n^2 there is a resonance and no eigenvalue. The same statements for C_n hold with the list $j = 1, \dots, n-1$.

Proof. The seed $\phi_{m,0} = \operatorname{sech}^m x$ satisfies $A_m \phi_{m,0} = 0$. For $0 \le j < m$, repeated use of (4) gives the nonzero state

$$\phi_{m,j} = A_m^* A_{m-1}^* \cdots A_{m-j+1}^* \operatorname{sech}^{m-j} x, \qquad B_m \phi_{m,j} = j(2m-j)\phi_{m,j}. \tag{6}$$

Its nodal degree, or equivalently the nonvanishing intertwining norms, shows that these states are simple and distinct.

Construction is not yet exhaustion. If a further L^2 eigenstate of B_m had energy below m^2, applying $A_m, A_{m-1}, \dots$ either lands in one of the one-dimensional kernels already used in (6), or produces an L^2 eigenstate below the continuum of $B_0 = -d^2/dx^2$. The latter has none. Reversing the intertwiners therefore leaves exactly (5).

It remains to prove the spectral type rather than infer it from formal Jost solutions. Put

$$D_m = A_m^* A_{m-1}^* \cdots A_1^*, \qquad B_0 = -d^2/dx^2, \qquad Q_m = \prod_{r=1}^{m} (B_0 + r^2).$$

Induction from (4), first on Schwartz functions and then by closure, gives

$$B_m D_m = D_m (B_0 + m^2), \qquad D_m^* D_m = Q_m. \tag{7}$$

Since $Q_m \ge (m!)^2$, the initially defined operator $U_m = D_m Q_m^{-1/2}$ extends to an isometry on $L^2(\mathbb{R})$ and satisfies $B_m U_m = U_m (B_0 + m^2)$. Its range is closed and equals the closure of $\operatorname{ran} D_m$; hence its orthogonal complement is $\ker D_m^*$.

Each of the m states in (6) belongs to this kernel: applying the adjoint intertwining would otherwise create an L^2 eigenstate of $B_0 + m^2$ below m^2. Conversely, D_m^* is an order-m differential operator, so its solution space, and therefore its L^2 kernel, has dimension at most m. Thus

$$(\operatorname{ran} U_m)^{\perp} = \operatorname{span}\{\phi_{m,0}, \dots, \phi_{m,m-1}\}.$$

The restriction of B_m to the orthogonal complement of these bound states is consequently unitarily equivalent to $B_0 + m^2$. This proves at once that its remaining spectrum is $[m^2, \infty)$, purely absolutely continuous, with no embedded eigenvalue or singular-continuous part. Finally, (4) transfers the complete statement to C_n. $\square$

3 Dirac point spectrum and chirality

The equations $A_n u = 0$ and $A_n^* v = 0$ give $u = c \operatorname{sech}^n x$ and $v = c' \cosh^n x$. Exactly the first is square-integrable, so $\ker H_n$ is one-dimensional and upper chiral. Its squared norm is

$$\int_{\mathbb{R}} \operatorname{sech}^{2n} x \, dx = \frac{4^n n!(n-1)!}{(2n)!}. \tag{8}$$

For every $q > 0$, A_n and A_n^* identify the q-eigenspaces of B_n and C_n. If $B_n u = qu$, the two normalized linear combinations of u and $q^{-1/2} A_n u$ have Dirac energies $\pm\sqrt{q}$. Lemma 1 now proves (2), including all multiplicities. The anticommutation of H_n with $\operatorname{diag}(1, -1)$ is the structural reason for the nonzero sign pairing.

4 Thresholds and reflectionless scattering

Start with the free plane wave e^{ikx}, $k > 0$, and set

$$\Psi_{m,k} = A_m^* A_{m-1}^* \cdots A_1^* e^{ikx}. \tag{9}$$

Intertwining gives $B_m \Psi_{m,k} = (m^2 + k^2)\Psi_{m,k}$. With the time convention e^{-iEt}, each factor tends to $-ik + r$ at $+\infty$ and to $-ik - r$ at $-\infty$. Thus

$$\Psi_{m,k} \sim C_- e^{ikx} \quad (x \to -\infty), \qquad \Psi_{m,k} \sim C_+ e^{ikx} \quad (x \to +\infty),$$

where $C_- = \prod_{r=1}^{m}(-r - ik)$ and $C_+ = \prod_{r=1}^{m}(r - ik)$. Normalizing the incoming coefficient at the left gives

$$r_m(k) = 0, \qquad t_m(k) = \frac{C_+}{C_-} = \prod_{r=1}^{m} \frac{k + ir}{k - ir}, \qquad |t_m(k)| = 1. \tag{10}$$

This equation fixes the phase convention; changing the Jost convention may invert t_m but cannot alter reflectionlessness. The lower channel is the adjacent member B_{m-1} plus a constant, so it is reflectionless too. For nonzero Dirac energy the other spinor component is recovered by a first-order application of A_n or A_n^*; no reflected exponential can be created. This proves assertion (v) without asserting a convention-independent Dirac transmission phase.

At $k = 0$, (9) becomes

$$F_m := \Psi_{m,0} = D_m 1, \qquad F_m(+\infty) = m!, \qquad F_m(-\infty) = (-1)^m m!.$$

Indeed, F_m is a polynomial in $\tanh x$ and $B_m F_m = m^2 F_m$; its endpoint values follow successively from the limiting coefficients $+r$ and $-r$. The threshold spinors are

$$\begin{pmatrix} F_m \\ (\pm m)^{-1} A_m F_m \end{pmatrix}.$$

Because $(A_m F_m)(+\infty) = m\, m!$ and $(A_m F_m)(-\infty) = (-1)^{m+1} m\, m!$, they are bounded and nonzero at both ends, hence not in L^2. The unitary decomposition (7) shows that the free threshold has no hidden L^2 preimage, so neither Dirac threshold is an eigenvalue. This proves assertion (iv).